

Cafe Tips Example

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Statistical Inference

Universe

- jazz: $n = 25$
- pop: $n = 25$

Variables

Symbol	Meaning	Type / Range
μ_1, μ_2	population means for groups 1 and 2	$\mathbb{R}$
$\bar{x}_{\{x\}}_1, \bar{x}_{\{x\}}_2$	sample means	$\mathbb{R}$
s_1, s_2	sample standard deviations	$\mathbb{R}_+$
n_1, n_2	sample sizes	$\mathbb{Z}^+$
t	test statistic (Welch's t-test)	$\mathbb{R}$
p	p-value	$[0, 1]$
α	significance level	$[0, 1]$
d	Cohen's d (effect size)	$\mathbb{R}_{\geq 0}$

Structure

- **Jazz days**: 25 days, mean tip 18.2% (std 2.2)
- **Pop days**: 25 days, mean tip 15.9% (std 3.0)
- **Question**: Does music genre significantly affect tips?

Real-World Mapping

Real-World Concept	Mathematical Object
group average	sample mean $\bar{x}$
group variability	sample std s
sample count	n
significance threshold	$\alpha = 0.05$

Constraints

(1) $H_0 : \mu_1 = \mu_2$

null hypothesis

(2) $H_1 : \mu_1 \neq \mu_2$

alternative hypothesis

(3) $\alpha = 0.05$

significance level

(4)
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's t-test statistic

Objective

Reject H_0 if $p < \alpha$

Approach

Suggested approach unknown

Complexity class:

Available tools: unknown

2. Solution

Answer:	$t = 3.0857$; $p = 0.003484$; significant
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	unknown
Time:	0.0000s
Certificate:	Assumptions checked; test statistic independently verified.

Solution Details

Descriptive Statistics:

jazz: $n = 25$, mean = 18.16, std = 2.2410934831014973

pop: $n = 25$, mean = 15.86, std = 2.977834783865619

Assumptions checked:

normality_jazz: PASS

normality_pop: PASS

equal_variance: PASS

Test: Welch's t-test

$t = 3.0857$

p-value = 0.003484

Significant at alpha = 0.05: Yes

Mean difference: 2.3000

Cohen's d = 0.8728 (large)

95% CI for difference: [0.7983, 3.8017]

Bayesian Analysis:

Power Analysis:

Recommendation: **STRONG EVIDENCE:** Jazz music significantly increases tips with a large effect. Switch to jazz during peak hours.

Verification

✓ All Agree

Passed

3. Interpretation

The Question

- **jazz days**: 25 days, mean tip 18.

The Answer

Statistically significant; $p = 0.0035$; $d = 0.87$ (large).

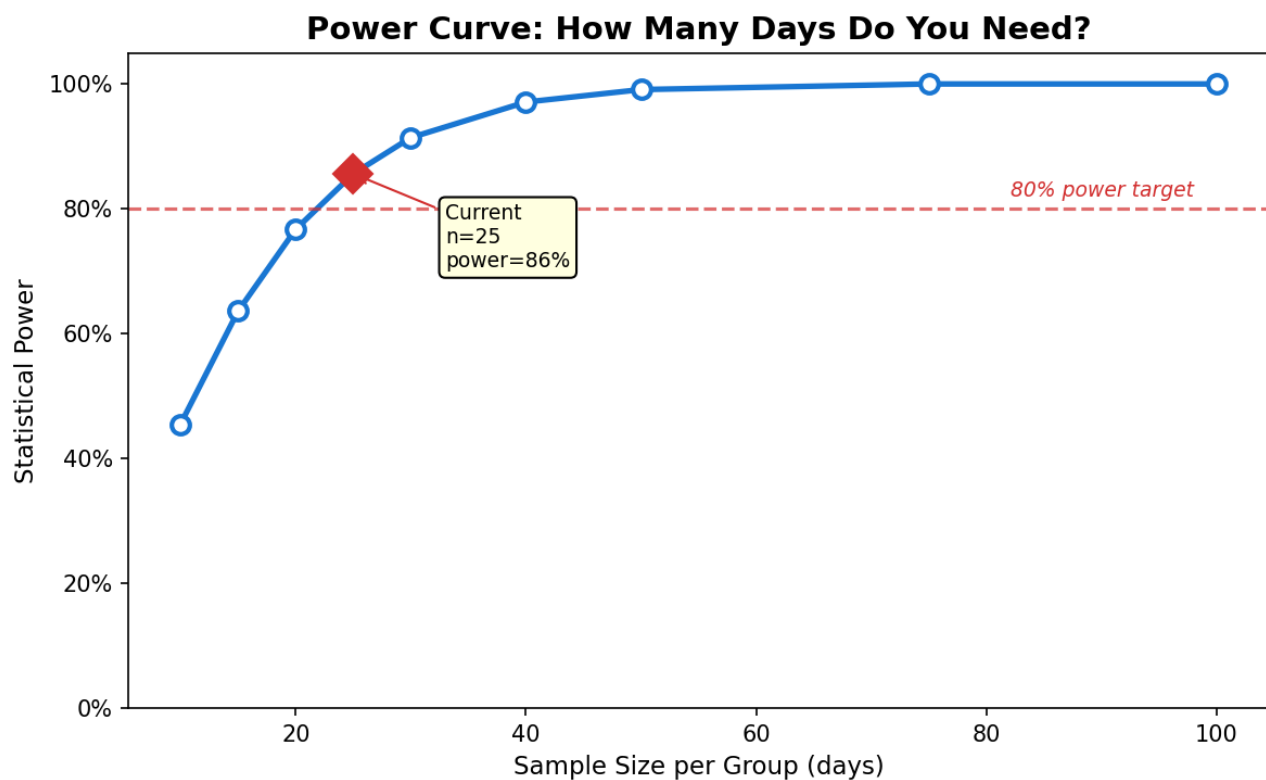
What This Means

- t-statistic: 3.09, p-value: 0.0035 (significant at $\alpha=0.05$ and 0.01)
- Cohen's d: 0.87 (large effect)
- Mean difference: \$2.30 higher tips on jazz days
- 95% CI: [\$0.80, \$3.80]
- $P(\text{jazz} > \text{pop})$: 99.9% (Bayesian)
- Power: 85.6% (well-powered)
- All 3 verification methods agree: Mann-Whitney, permutation test, bootstrap CI

Sensitivity Analysis

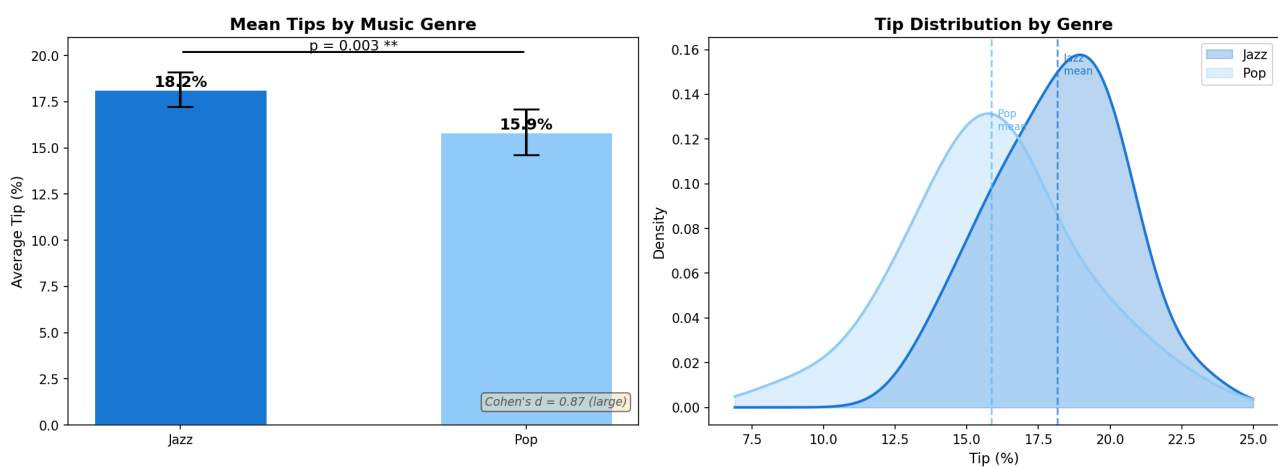
Parameter	Current	Change	New Obj.	Class
power_curve	present	removed		robust
stricter_alpha	present	removed		robust

Visualizations



Cafe Power Curve

Cafe Tips Analysis: Jazz vs. Pop Music



Cafe Tips Analysis

Recommendations

1. Report effect sizes alongside p-values for practical significance assessment.

Limitations

- Results are conditional on model assumptions (normality, independence, etc.).